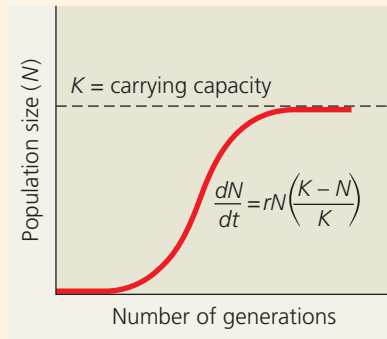


CONCEPT 53.3

The logistic model describes how a population grows more slowly as it nears its carrying capacity

(pp. 1197–1200)

- Exponential growth cannot be sustained in any population. A more realistic population model limits growth by incorporating **carrying capacity** (K), the maximum population size the environment can support.
- According to the **logistic growth** equation $dN/dt = rN(K - N)/K$, growth levels off as population size approaches the carrying capacity.



- The logistic model fits few real populations perfectly, but it is useful for estimating possible growth.

? As an ecologist who manages a wildlife preserve, you want to increase the preserve's carrying capacity for a particular endangered species. How might you go about accomplishing this?

CONCEPT 53.4

Life history traits are products of natural selection

(pp. 1200–1202)

- Life history** traits are evolutionary outcomes reflected in the development, physiology, and behavior of organisms.
- Big-bang, or **semelparous**, organisms reproduce once and die. **Iteroparous** organisms produce offspring repeatedly.
- Life history traits such as brood size, age at maturity, and parental caregiving represent trade-offs between conflicting demands for time, energy, and nutrients. Two hypothetical life history patterns are **K-selection** and **r-selection**.

? Explain why ecological trade-offs are common.

CONCEPT 53.5

Density-dependent factors regulate population growth (pp. 1202–1207)

- In **density-dependent** population regulation, death rates rise and birth rates fall with increasing density. A birth or death rate that does not vary with density is said to be **density independent**.
- Density-dependent changes in birth and death rates curb population increase through negative feedback and can eventually stabilize a population near its carrying capacity. Density-dependent limiting factors include intraspecific competition for limited food or space, increased predation, disease, and intrinsic physiological factors.
- Because changing environmental conditions periodically disrupt them, all populations exhibit some size fluctuations. Many

populations undergo regular boom-and-bust cycles that are influenced by complex interactions between biotic and abiotic factors. A **metapopulation** is a group of populations linked by immigration and emigration.

? Give an example of one biotic and one abiotic factor that contribute to yearly fluctuations in the size of the human population.

CONCEPT 53.6

The human population is no longer growing exponentially but is still increasing extremely rapidly (pp. 1207–1211)

- Since about 1650, the global human population has grown exponentially, but within the last 50 years, the rate of growth has fallen by half. Differences in **age structure** show that while some nations' populations are growing rapidly, those of others are stable or declining in size. Infant mortality rates and life expectancy at birth vary widely in different countries.
- Ecological footprint** is the aggregate land and water area needed to produce all the resources a person or group of people consume and to absorb all of their waste. It is one measure of how close we are to the carrying capacity of Earth, which is uncertain. With a world population of more than 7.6 billion people, we are already using many resources in an unsustainable manner.

? How do humans differ from other species in the ability to "choose" a carrying capacity for their environment?

TEST YOUR UNDERSTANDING

➔ For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

- Population ecologists follow the fate of same-age cohorts to
 - determine a population's carrying capacity.
 - determine the birth rate and death rate of each group in a population.
 - determine if a population is regulated by density-dependent processes.
 - determine the factors that affect the size of a population.
- A population's carrying capacity
 - may change as environmental conditions change.
 - can be accurately calculated using the logistic growth model.
 - increases as the per capita population growth rate decreases.
 - can never be exceeded.
- Scientific study of the population cycles of the snowshoe hare and its predator, the lynx, has revealed that
 - predation is the dominant factor affecting prey population cycling.
 - hares and lynx are so mutually dependent that each species cannot survive without the other.
 - both hare and lynx population sizes are affected mainly by abiotic factors.
 - the hare population is *r*-selected and the lynx population is *K*-selected.